

Title

Protocol for a cluster randomized trial of behaviour modification integrated with cardiac rehabilitation to sustain exercise adherence

Abstract

Background: Sustaining physical activity gains achieved during cardiac rehabilitation programs remains difficult. Standard programs often fail to produce lasting behavioural change. We describe a protocol for a cluster-randomized trial testing whether targeted behaviour-change coaching can enhance and prolong physical activity participation in patients completing coronary rehabilitation. **Methods:** Six community-based cardiac rehabilitation clinics will participate. Approximately 96 individuals in the maintenance phase of cardiac rehabilitation will be cluster-randomized to receive either standard care or standard care plus a structured 6-session behaviour modification program delivered by trained facilitators. Randomization occurs at the clinic level to reduce contamination. **Primary outcome:** objectively measured daily moderate-to-vigorous physical activity and step counts at 6 months, assessed via wearable accelerometer. **Secondary outcomes include** self-reported health status, life satisfaction indices, and psychological well-being. **Eligibility:** clinically stable coronary patients with ejection fraction >40%. **Analysis** will employ mixed models accounting for cluster design. This trial aims to identify scalable strategies for maintaining cardiovascular exercise participation.